

Operating System Lab
Lab Task 7



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Section 4A

Question 01

```
MariaDB [lab_09_task]> SELECT phone_id, model_name, rating
-> FROM smartphones
-> WHERE rating=(
-> SELECT MAX(rating)
-> FROM smartphones
-> WHERE rating<(
-> SELECT MAX(rating) FROM smartphones
-> )
-> );
```

phone_id	model_name	rating
3	Galaxy S23	4.7
8	Pixel 7	4.7

2 rows in set (0.001 sec)

```
MariaDB [lab_09_task]> |
```

Question 02

```
MariaDB [lab_09_task]> SELECT phone_id, model_name, brand_id, price
-> FROM smartphones
-> WHERE brand_id IN (
->     SELECT brand_id FROM brands WHERE country = 'USA'
-> );
```

phone_id	model_name	brand_id	price
1	iPhone 14	1	999.00
2	iPhone 13	1	799.00
8	Pixel 7	5	799.00

3 rows in set (0.001 sec)

```
MariaDB [lab_09_task]> |
```

Question 03

```
MariaDB [lab_09_task]> SELECT customer_id, name, city, country
-> FROM customers
-> WHERE customer_id IN (
->     SELECT customer_id FROM orders
-> );
```

customer_id	name	city	country
1	Ali Khan	Lahore	Pakistan
2	Sara Ahmed	Karachi	Pakistan
3	John Smith	London	UK

3 rows in set (0.001 sec)

Question 04

```
MariaDB [lab_09_task]> SELECT phone_id, model_name, brand_id, price
-> FROM smartphones s1
-> WHERE price = (
-> SELECT MIN(price)
-> FROM smartphones s2
-> WHERE s2.brand_id = s1.brand_id
-> );
```

phone_id	model_name	brand_id	price
2	iPhone 13	1	799.00
4	Galaxy A54	2	499.00
5	Redmi Note 12	3	299.00
7	OnePlus 11	4	699.00
8	Pixel 7	5	799.00

5 rows in set (0.001 sec)

```
MariaDB [lab_09_task]> |
```

Question 05

```
MariaDB [lab_09_task]> SELECT customer_id, name, city
-> FROM customers c
-> WHERE NOT EXISTS (
-> SELECT 1 FROM reviews r WHERE r.customer_id = c.customer_id
-> );
```

customer_id	name	city
3	John Smith	London

1 row in set (0.001 sec)

```
MariaDB [lab_09_task]> |
```

QUESTION 06

```

MariaDB [lab_09_task]> SELECT phone_id, model_name, release_year, battery_capacity
-> FROM smartphones
-> WHERE battery_capacity > ALL (
-> SELECT battery_capacity
-> FROM smartphones
-> WHERE release_year < 2022
-> );

```

phone_id	model_name	release_year	battery_capacity
4	Galaxy A54	2023	5000
5	Redmi Note 12	2023	5000
7	OnePlus 11	2023	5000

```

3 rows in set (0.001 sec)

MariaDB [lab_09_task]> |

```

QUESTION 07

```

MariaDB [lab_09_task]> SELECT
-> s1.phone_id,
-> s1.model_name,
-> s1.brand_id,
-> s1.rating AS phone_rating, (
-> SELECT ROUND(AVG(s2.rating), 2
-> )
-> FROM smartphones s2
-> WHERE s2.brand_id = s1.brand_id
-> )
-> AS avg_brand_rating
-> FROM smartphones s1;

```

phone_id	model_name	brand_id	phone_rating	avg_brand_rating
1	iPhone 14	1	4.8	4.70
2	iPhone 13	1	4.6	4.70
3	Galaxy S23	2	4.7	4.50
4	Galaxy A54	2	4.3	4.50
5	Redmi Note 12	3	4.2	4.30
6	Mi 11	3	4.4	4.30
7	OnePlus 11	4	4.5	4.50
8	Pixel 7	5	4.7	4.70

```

8 rows in set (0.001 sec)

MariaDB [lab_09_task]> |

```

QUESTION 08

```
MariaDB [lab_09_task]> SELECT
  -> c.customer_id,
  -> c.name, (
  -> SELECT COUNT(*)
  -> FROM orders o
  -> WHERE o.customer_id = c.customer_id
  -> )
  -> AS total_orders
  -> FROM customers c;
```

customer_id	name	total_orders
1	Ali Khan	2
2	Sara Ahmed	1
3	John Smith	1
4	Maria Garcia	0

```
4 rows in set (0.001 sec)

MariaDB [lab_09_task]> |
```

QUESTION 09

```
MariaDB [lab_09_task]> UPDATE smartphones
  -> SET price = price * 0.80
  -> WHERE rating < (
  -> SELECT AVG(rating) FROM (SELECT rating FROM smartphones) AS temp
  -> );
Query OK, 4 rows affected (0.019 sec)
Rows matched: 4 Changed: 4 Warnings: 0

MariaDB [lab_09_task]> |
```

QUESTION 10

```
MariaDB [lab_09_task]> DELETE FROM smartphones
  -> WHERE phone_id NOT IN (
  -> SELECT phone_id FROM (SELECT phone_id FROM order_items) AS temp
  -> );
Query OK, 3 rows affected (0.018 sec)

MariaDB [lab_09_task]> |
```

NOTE:

Last 2 queries are not working tried multiple time
unable to figure out the problem even after assisting from AI.